

Functional Equations — Solutions

Problem 1. (Peru TST 2012 Problem 1) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that

$$f(f(x)) = \frac{x^2 - x}{2} \cdot f(x) + 2 - x, \quad (1)$$

for all $x \in \mathbb{R}$. Find all possible values of $f(2)$.

Solution. Answer: $f(2) = 2$

Putting $x = 0$ and $x = 2$ in (1) respectively, we obtain

$$f(f(0)) = 2, \quad (2)$$

$$f(f(2)) = f(2). \quad (3)$$

Putting $x = f(2)$ in (1), we find that

$$f^3(2) = \frac{f(2)^2 - f(2)}{2} \cdot f(f(2)) + 2 - f(2).$$

Letting $c = f(2)$ and using (3), this becomes $c = \frac{c^2 - c}{2} \cdot c + 2 - c$. This can be rewritten as $c^3 - c^2 - 4c + 4 = 0$, i.e. $(c - 1)(c - 2)(c + 2) = 0$. This implies $c = 2, 1, -2$.

Putting $x = f(0)$ in (1), we find that

$$f^3(0) = \frac{f(0)^2 - f(0)}{2} \cdot f(f(0)) + 2 - f(0).$$

Using (2), this becomes

$$c = f(0)^2 - 2f(0) + 2 = (f(0) - 1)^2 + 1. \quad (4)$$

This shows $c \geq 1$, and so $c = -2$ is rejected.

If $c = 1$, by (3) we have $f(1) = 1$. Also, by (4) we have $f(0) = 1$. But then (2) becomes $f(1) = 2$, contradicting $f(1) = 1$.

Thus, the only possibility is $c = 2$. This is possible. For example, consider

$$f(x) = \begin{cases} \frac{4}{x} & \text{if } x \neq 0, \\ 2 & \text{if } x = 0. \end{cases}$$

We claim that it satisfies (1). Indeed, if $x \neq 0$, then $f(x) = \frac{4}{x} \neq 0$, and so

$$f(f(x)) = x = \frac{x^2 - x}{2} \cdot \frac{4}{x} + 2 - x = \frac{x^2 - x}{2} \cdot f(x) + 2 - x.$$

If $x = 0$, then

$$f(f(0)) = f(2) = 2 = \frac{0^2 - 0}{2} \cdot f(0) + 2 - 0.$$

In this case, $f(2) = 2$. □

Problem 2. (Czech and Slovak Olympiad 2017 Round III Category A Problem 3) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(y - xy) = f(x)y + (x - 1)^2 f(y) \quad (1)$$

holds for all real numbers x, y .

Solution. Answer: $f(x) = cx(x - 1)$ where c is a constant

Putting $x = 1$ in (1), we obtain $f(0) = f(1)y$ for any $y \in \mathbb{R}$. Thus, we must have $f(0) = f(1) = 0$. Next, putting $y = 1$ in (1), we obtain

$$f(1 - x) = f(x) \quad (2)$$

Now, replacing x by $1 - x$ in (1) and using (2), we find that

$$f(xy) = f(x)y + x^2 f(y).$$

Swapping x and y and comparing with this equation, we get $f(x)y + x^2 f(y) = f(y)x + y^2 f(x)$. For $x, y \neq 0, 1$, this is the same as

$$\frac{f(x)}{x(x - 1)} = \frac{f(y)}{y(y - 1)}.$$

Thus, this is a constant, and so $f(x) = cx(x - 1)$ for some constant c . Note that this also holds when $x = 0, 1$. It remains to check that all such functions are solutions. Indeed,

$$\begin{aligned} f(y - xy) &= c(y - xy)(y - xy - 1) \\ &= cy(1 - x)(y - xy - 1), \\ f(x)y + (x - 1)^2 f(y) &= cx(x - 1)y + (x - 1)^2 cy(y - 1) \\ &= cy(x - 1)[x + (x - 1)(y - 1)] \\ &= cy(1 - x)(y - xy - 1). \end{aligned}$$

The two sides are equal. □

Problem 3. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2 + f(y)) = (x - y)^2 f(x + y) \quad (1)$$

for all real numbers x and y .

Solution. Answer: $f(x) = 0$ for all $x \in \mathbb{R}$ or $f(x) = -x^2$ for all $x \in \mathbb{R}$

Putting $y = 0$ in (1), we obtain

$$f(x^2 + f(0)) = x^2 f(x).$$

Replacing x by $-x$ and comparing with this equation, we get $x^2 f(x) = x^2 f(-x)$. This implies

$$f(x) = f(-x) \quad (2)$$

for any $x \in \mathbb{R}$ (which also holds when $x = 0$). Replacing y by $-y$ in (1), we see that

$$f(x^2 + f(-y)) = (x + y)^2 f(x - y). \quad (3)$$

Note that $f(x^2 + f(-y)) = f(x^2 + f(y))$ by (2). Therefore, comparing (1) and (3), we get

$$(x - y)^2 f(x + y) = (x + y)^2 f(x - y).$$

Using the change of variables $u = x - y$ and $v = x + y$, we have $u^2 f(v) = v^2 f(u)$. Note that u and v can be any real numbers. Thus, we have

$$\frac{f(u)}{u^2} = \frac{f(v)}{v^2}$$

for any nonzero real numbers u, v . This implies $f(u) = cu^2$ for all $u \neq 0$ for some constant c . Putting $x = y = 0$ in (1), we obtain $f(f(0)) = 0$. If $f(0) \neq 0$, then this yields $cf(0)^2 = 0$, and hence $c = 0$. But then by putting $x = -1$ and $y = 1$ in (1), we get the contradiction $0 = f(0)$. Therefore, $f(0) = 0$. Thus, $f(x) = cx^2$ holds for all $x \in \mathbb{R}$.

Now, putting $x = y$ in (1), we obtain $f(x^2 + f(x)) = 0$. This implies $c(x^2 + cx^2)^2 = 0$, and hence $c(c + 1)^2 x^4 = 0$. Therefore, we must have $c = -1, 0$. In other words, $f(x) = -x^2$ for all x or $f(x) = 0$ for all x . The zero function is clearly a solution. For $f(x) = -x^2$, we check that

$$f(x^2 + f(y)) = -(x^2 - y^2)^2 = -(x - y)^2(x + y)^2 = (x - y)^2 f(x + y).$$

So both are solutions. □

Problem 4. Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f\left(\frac{x + y}{2}\right) = \frac{2f(x)f(y)}{f(x) + f(y)} \quad (1)$$

for all positive real numbers x and y .

Solution. Answer: $f(x) = \frac{1}{c + xd}$ for some constants $c, d > 0$

Let $g(x) = \frac{1}{f(x)}$. The equation (1) can be rewritten as $\frac{1}{f(x)} + \frac{1}{f(y)} = \frac{2}{f\left(\frac{x+y}{2}\right)}$. This means

$$\frac{g(x) + g(y)}{2} = g\left(\frac{x + y}{2}\right).$$

This shows the image of an arithmetic sequence is an arithmetic sequence. In particular, by letting $g(2) - g(1) = d$ and $g(1) = c + d$, we easily obtain $g(n) = c + nd$ for all $n \in \mathbb{N}$.

Next, for any $n \in \mathbb{N}$, we consider the arithmetic sequence $\frac{1}{n}, \frac{2}{n}, \dots$. Since 1 and 2 are two terms of this arithmetic sequence, we easily deduce

$$g\left(\frac{m+1}{n}\right) - g\left(\frac{m}{n}\right) = \frac{1}{n}[g(2) - g(1)] = \frac{d}{n}$$

for any $m \in \mathbb{N}$. Also, since $g(1) = c + d$, it is not hard to find that $g\left(\frac{m}{n}\right) = c + \frac{m}{n} \cdot d$ for any $m \in \mathbb{N}$. Thus, $g(r) = c + rd$ for any $r \in \mathbb{Q}^+$. Note that $c, d > 0$ since $g(r) > 0$ for all $r \in \mathbb{Q}^+$.

Lastly, for any $x \in \mathbb{R}^+$ and $n \in \mathbb{N}$, there exists $r \in \mathbb{Q}^+$ such that $x - \frac{x}{n} < r < x$. By considering the arithmetic sequence $x, x - (x - r), x - 2(x - r), \dots, x - (n - 1)(x - r)$ (note that the last term is positive since $x - (n - 1)(x - r) > x - \frac{n - 1}{n}x > 0$), we see that

$$g(x - (n - 1)(x - r)) = g(x) - (n - 1)[g(x) - g(r)] = (n - 1)(c + rd) - (n - 2)g(x).$$

Since $g(x - (n - 1)(x - r)) > 0$, this implies $g(x) < \frac{(n - 1)(c + rd)}{n - 2}$. When n tends to infinity, since $r \rightarrow x$, this yields

$$g(x) \leq c + xd.$$

Similarly, there exists $r' \in \mathbb{Q}^+$ such that $x < r' < x + \frac{x}{n}$. Using similar arguments, we find that

$$g(r' - (n - 1)(r' - x)) = g(r') - (n - 1)[g(r') - g(x)] = (n - 1)g(x) - (n - 2)(c + r'd),$$

and hence $g(x) > \frac{(n - 2)(c + r'd)}{n - 1}$. When n tends to infinity, this yields

$$g(x) \geq c + xd.$$

Thus, we must have $g(x) = c + xd$ for any $x \in \mathbb{R}^+$. We check that all such functions are solutions. Indeed,

$$\frac{g(x) + g(y)}{2} = \frac{c + xd + c + yd}{2} = c + \frac{x + y}{2} \cdot d = g\left(\frac{x + y}{2}\right).$$

So we have $f(x) = \frac{1}{c + xd}$ for some constants $c, d > 0$. □

Problem 5. (Italy TST 2007 Problem 3) Find all $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xy + f(x)) = xf(y) + f(x) \tag{1}$$

for every pair of real numbers x, y .

Solution. Answer: $f(x) = 0$ for all $x \in \mathbb{R}$ or $f(x) = x$ for all $x \in \mathbb{R}$

Clearly, the zero function is a solution. We now assume f is not the zero function.

Claim 1. If $f(a) = f(b) \neq 0$, then $a = b$.

Proof. Suppose $f(a) = f(b) \neq 0$. By putting $x = a$ and $y = b$ in (1), we obtain

$$f(ab + f(a)) = af(b) + f(a).$$

Swapping a and b , we get

$$f(ab + f(b)) = bf(a) + f(b).$$

Since $f(a) = f(b)$, the two expressions on the left are equal. Thus, we find that

$$af(b) + f(a) = bf(a) + f(b).$$

This implies $(a - 1)f(b) = (b - 1)f(a)$, and hence $a = b$ since $f(b) = f(a) \neq 0$. $\square$

Claim 2. If $f(a) = 0$, then $a = 0$.

Proof. Suppose on the contrary that $f(a) = 0$ for some $a \neq 0$. Putting $x = a$ and $y = 1$ in (1), we obtain $0 = af(1)$. This implies $f(1) = 0$. Putting $y = 1$ in (1), we get

$$f(x + f(x)) = f(x).$$

Now, there exists x such that $f(x) \neq 0$ by assumption. For this x , the above equation becomes $x + f(x) = x$ by Claim 1. But this contradicts $f(x) \neq 0$. Thus, we must have $a = 0$ if $f(a) = 0$. $\square$

Now, Claim 1 and Claim 2 imply f is injective. Putting $x = y = 0$ in (1), we have $f(f(0)) = f(0)$, and hence $f(0) = 0$. Also, putting $y = 0$ in (1), we have $f(f(x)) = f(x)$, so that $f(x) = x$. It is easy to check that the identity function is another solution. $\square$

Problem 6. Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f\left(\frac{f(x)}{y}\right) = yf(y)f(f(x)) \quad (1)$$

for all positive real numbers x and y .

Solution. Answer: $f(x) = \frac{1}{\sqrt{x}}$

Putting $y = 1$ in (1), we get $f(1) = 1$. Putting $x = 1$ in (1), we get

$$f\left(\frac{1}{y}\right) = yf(y). \quad (2)$$

Also, putting $y = f(x)$ in (1), we get $1 = f(x)f(f(x))^2$. This implies

$$f(f(x)) = \frac{1}{\sqrt{f(x)}}. \quad (3)$$

Next, putting $y = \frac{1}{x}$ in (1) and using (2) and (3), we find that

$$f(xf(x)) = \frac{1}{x}f\left(\frac{1}{x}\right)f(f(x)) = \frac{1}{x}(xf(x))\frac{1}{\sqrt{f(x)}} = \sqrt{f(x)}.$$

By (2), this becomes

$$f\left(f\left(\frac{1}{x}\right)\right) = \sqrt{f(x)}. \quad (4)$$

Now, replacing x by $\frac{1}{x}$ in (3), we have

$$f\left(f\left(\frac{1}{x}\right)\right) = f\left(\frac{1}{x}\right)^{-\frac{1}{2}}. \quad (5)$$

Comparing (4) and (5), we obtain

$$f\left(\frac{1}{x}\right) = \frac{1}{f(x)}.$$

Comparing this with (2), we finally obtain $xf(x)^2 = 1$, i.e. $f(x) = \frac{1}{\sqrt{x}}$. We check that

$$f\left(\frac{f(x)}{y}\right) = f\left(\frac{1}{y\sqrt{x}}\right) = \sqrt{y\sqrt{x}} = \sqrt{y}\sqrt[4]{x}$$

and

$$yf(y)f(f(x)) = y \cdot \frac{1}{\sqrt{y}}f\left(\frac{1}{\sqrt{x}}\right) = \sqrt{y}\sqrt[4]{x}$$

are equal. □

Problem 7. (Japan MO 2016 Problem 4) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(yf(x) - x) = f(x)f(y) + 2x \quad (1)$$

for all $x, y \in \mathbb{R}$.

Solution. Answer: $f(x) = -2x$ for all $x \in \mathbb{R}$ or $f(x) = 1 - x$ for all $x \in \mathbb{R}$

Putting $x = y = 0$ in (1), we obtain $f(0) = f(0)^2$. So $f(0) = 0, 1$.

If $f(0) = 0$, putting $y = 0$ in (1), we get $f(-x) = 2x$. This implies $f(x) = -2x$ for all $x \in \mathbb{R}$. We check that

$$f(yf(x) - x) = -2(-2xy - x) = 4xy + 2x = (-2x)(-2y) + 2x = f(x)f(y) + 2x.$$

If $f(0) = 1$, putting $y = 0$ in (1), we get

$$f(-x) = f(x) + 2x. \quad (2)$$

Replacing y by $-y$ in (1), we have

$$f(-yf(x) - x) = f(x)f(-y) + 2x.$$

Using (2), this becomes

$$f(yf(x) + x) + 2(yf(x) + x) = f(x)(f(y) + 2y) + 2x.$$

This can be simplified as

$$f(yf(x) + x) = f(x)f(y).$$

Combining this with (1), we obtain

$$f(yf(x) - x) = f(yf(x) + x) + 2x.$$

Therefore, whenever $f(x) \neq 0$, we can replace y by $\frac{x}{f(x)}$ to get

$$f(2x) = 1 - 2x.$$

In other words, for any $x \in \mathbb{R}$, we must have $f(x) = 0$ or $f(2x) = 1 - 2x$.

Next, for those $x \in \mathbb{R}$ such that $f(x) = 0$, by (2) we have $f(-x) = 2x \neq 0$ since $x \neq 0$. It follows that $f(-2x) = 1 - (-2x) = 1 + 2x$, and hence $f(2x) = f(-2x) - 4x = 1 - 2x$. Therefore, $f(2x) = 1 - 2x$ holds for all $x \in \mathbb{R}$. This means $f(x) = 1 - x$ for all $x \in \mathbb{R}$. We check that

$$f(yf(x) - x) = 1 - (y(1 - x) - x) = xy + x - y + 1 = (1 - x)(1 - y) + 2x = f(x)f(y) + 2x.$$

Thus, the two solutions are $f(x) = -2x$ for all $x \in \mathbb{R}$ and $f(x) = 1 - x$ for all $x \in \mathbb{R}$ □

Problem 8. (Vietnam MO 2017 Problem 5) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying the relation:

$$f(xf(y) - f(x)) = 2f(x) + xy \quad \forall x, y \in \mathbb{R}. \tag{1}$$

Solution. Answer: $f(x) = 1 - x$

Putting $x = 1$ in (1), we obtain

$$f(f(y) - f(1)) = 2f(1) + y.$$

If $f(a) = f(b)$, then $a = f(f(a) - f(1)) - 2f(1) = f(f(b) - f(1)) - 2f(1) = b$, so f is injective. Also, replacing y by $y - 2f(1)$, we see that every $y \in \mathbb{R}$ is the image of something. Therefore, f is surjective.

Let t be such that $f(t) = 0$. Putting $x = t$ and $y = 0$ in (1), we get $f(tf(0)) = 0$. By the injectivity, this yields $tf(0) = t$. Thus, $t = 0$ or $f(0) = 1$. This means $f(0) = 0$ or $f(0) = 1$.

If $f(0) = 0$, putting $y = 0$ in (1), we have $f(-f(x)) = 2f(x)$. Since f is surjective, this implies $f(-y) = 2y$ for any $y \in \mathbb{R}$, i.e. $f(y) = -2y$ for any $y \in \mathbb{R}$. But then we check that

$$f(xf(y) - f(x)) = -2(-2xy + 2x) = 4xy - 4x$$

and

$$2f(x) + xy = -4x + xy$$

are not the same when $x, y \neq 0$.

Thus, we must have $f(0) = 1$. Putting $x = y = 1$ in (1), we obtain $1 = 2f(1) + 1$, and so $f(1) = 0$. Putting $x = 0$ in (1), we have $f(-1) = 2$. Also, putting $x = 1$ and $y = -1$ in (1), we find that $f(2) = -1$. Now, we put $y = 2$ in (1). This yields

$$f(-x - f(x)) = 2f(x) + 2x. \quad (2)$$

For any $x \in \mathbb{R}$, there exists $z \in \mathbb{R}$ such that $f(z) = x + f(x)$ since f is surjective. We put $x = z$ and $y = 1$ in (1). This gives

$$f(-x - f(x)) = 2x + 2f(x) + z. \quad (3)$$

Comparing (2) and (3), we plainly obtain $z = 0$. In other words, we have $x + f(x) = f(0) = 1$. Thus, $f(x) = 1 - x$ for any $x \in \mathbb{R}$. We check that

$$f(xf(y) - f(x)) = 1 - (x(1 - y) - (1 - x)) = 2 - 2x + xy = 2f(x) + xy.$$

□